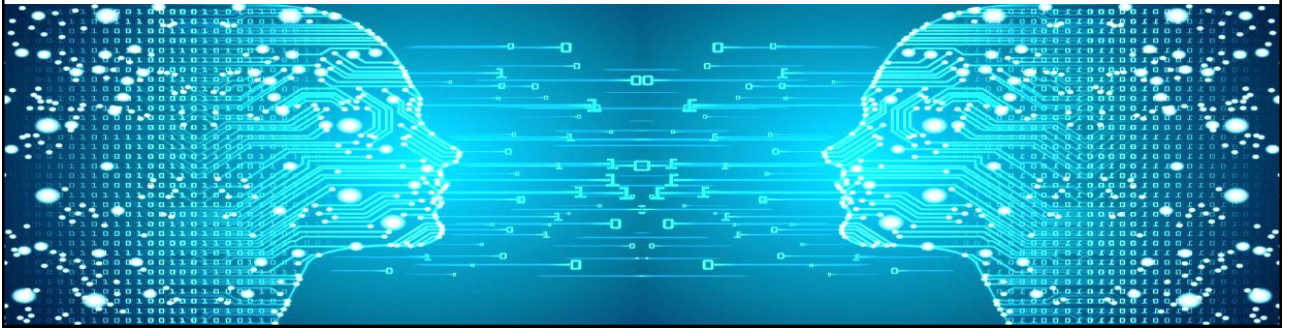
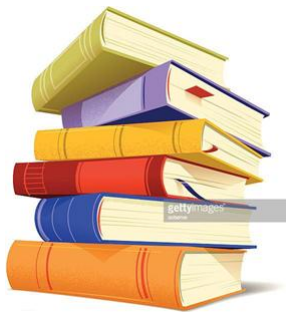


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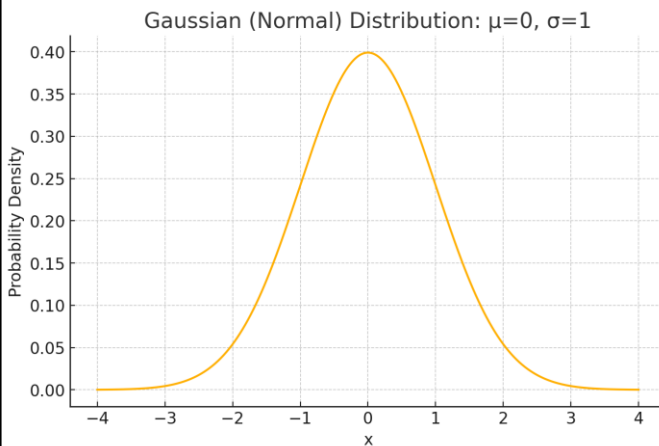


Gaussian Mixture Model - GMM



- Gaussian Distribution
- Gaussian Mixture Model
- Real-world Applications
- GMM Clustering
- Methods to Determine Number of Clusters (k)

What is a Gaussian Distribution?



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- **Gaussian = Normal distribution**
- Bell-shaped curve
- Symmetrical around μ
- Unimodal (only 1 peak)
- Defined by mean μ (mu) and variance σ^2 (sigma squared) (spread of the data)
- Appears naturally in many real-world phenomena
- Extremely important in statistics and machine learning

What is the 68–95–99.7–99.99 Rule?

- For the Gaussian (Normal) Distribution, there is a very famous rule:

Range around μ	Percentage of Data	Meaning
$\pm 1\sigma$	~68%	68% of the data lies within 1 standard deviation
$\pm 2\sigma$	~95%	95% of the data lies within 2 standard deviations
$\pm 3\sigma$	~99.7%	99.7% of the data lies within 3 standard deviations
$\pm 4\sigma$	~99.99%	99.99% of the data lies within 4 standard deviations

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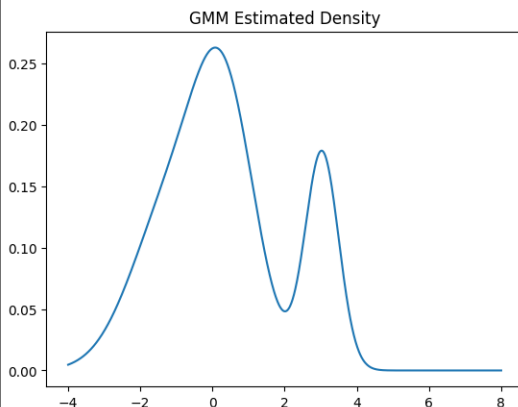
The probability density function of Gaussian

- The probability density function (PDF) of a Gaussian (Normal) distribution is defined as follows:

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Meaning:
 - The closer a value is to μ , the higher its probability
 - The farther a value is from μ , the faster the probability decreases

Limitation of Single Gaussian



- Real-world data often has **multiple peaks**.
- Single Gaussian cannot capture multi-modal structure.

What is GMM?

- Gaussian Mixture Models (GMM) are probabilistic models that represent data as a mixture of several Gaussian distributions.
- PDF of GMM = A weighted sum of Gaussian distributions:

$$p(x) = \sum_{k=1}^K w_k \mathcal{N}(x | \mu_k, \Sigma_k)$$

- Where:
 - (w_k) : weight of component k
 - (μ_k, Σ_k) : mean & covariance of Gaussian k
 - $\mathcal{N}(x | \mu_k, \Sigma_k)$: PDF of the k-th Gaussian distribution at point x
 - $\sum_{k=1}^K w_k = 1$

Real-World Applications

- Soft clustering
- Speech & audio processing
- Image processing & computer vision
- Finance & market modeling
- Customer segmentation
- Anomaly detection
- Density estimation

Hard vs. Soft clustering

- **Hard clustering:**
 - Every point belongs to one and only one cluster.
 - Example: K-means
 - Membership is binary: the point is either in a cluster or not.
 - Fast and simple, but cannot express uncertainty.
- **Soft clustering:**
 - Each point belongs to all clusters with different probabilities.
 - Example: GMM
 - Membership is a distribution across clusters (e.g., 0.7 in cluster A, 0.3 in cluster B).
 - Captures uncertainty and overlapping cluster boundaries.
 - More flexible and often more realistic for real-world data.

Gaussian Mixture Model Clustering

1. Initialize the parameters (w_k, μ_k, Σ_k)
2. E-step: compute **responsibility** (γ_{ik}) (the probability that point x_i belongs to Gaussian k)
3. M-step: update (w_k, μ_k, Σ_k)
4. Repeat the E-M steps until convergence
5. Assign clusters

Determining Number of Clusters (k)

- Common methods:
 - BIC (Bayesian Information Criterion)
 - AIC (Akaike Information Criterion)
 - Cross-validation
 - ICL (Integrated Completed Likelihood)
 - Dirichlet Process GMM (infinite mixtures)

BIC and AIC

- Choose K that minimizes:
 - $BIC = -2 \log L + k \log n$
 - $AIC = -2 \log L + 2k$
- Where:
 - L: likelihood function of the GMM model
 - n: number of observations in the data
 - k: number of clusters

Summary

- Today we learned:
 - What Gaussian and GMM are
 - Why GMM models multi-modal data
 - Real-world applications
 - EM algorithm
 - How clustering works in GMM
 - How to choose K

